

WEEK – 01

Merge Sort:

```
#include <stdio.h>
```

```
#include <time.h>
```

```
void merge(int arr[], int low, int mid, int high)
```

```
{
```

```
    int i = low, j = mid + 1, k = low;
```

```
    int temp[1000];
```

```
    while (i <= mid && j <= high)
```

```
    {
```

```
        if (arr[i] < arr[j])
```

```
            temp[k++] = arr[i++];
```

```
        else
```

```
            temp[k++] = arr[j++];
```

```
    }
```

```
    while (i <= mid)
```

```
        temp[k++] = arr[i++];
```

```
    while (j <= high)
```

```
        temp[k++] = arr[j++];
```

```
    for (i = low; i <= high; i++)
```

```
        arr[i] = temp[i];
```

```
void mergeSort(int arr[], int low, int high)
```

```
{
```

```
    if (low < high)
```

```
    {
```

```
        int mid = (low + high) / 2;
```

```
        mergeSort(arr, low, mid);
        mergeSort(arr, mid + 1, high);
        merge(arr, low, mid, high);
    }
}
```

```
int main()
{
    int n, i;
    int arr[1000];
    clock_t start, end;
    double time_taken;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d numbers:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    start = clock();

    mergeSort(arr, 0, n - 1);

    end = clock();

    printf("\nSorted elements are:\n");
    for (i = 0; i < n; i++)
        printf("%d ", arr[i]);

    time_taken = (double)(end - start) / CLOCKS_PER_SEC;
```

```
printf("\n\nTime taken to sort: %f seconds\n", time_taken);

return 0;

}
```

Output:

```
PS C:\Users\SHALINI K A\Desktop\ADA> gcc MergeSort.c -o MergeSort
PS C:\Users\SHALINI K A\Desktop\ADA> ./MergeSort
Enter number of elements: 6
PS C:\Users\SHALINI K A\Desktop\ADA> ./MergeSort
Enter number of elements: 6
Enter number of elements: 6
Enter 6 numbers:
23
10
5
89
46
45

Sorted elements are:
5 10 23 45 46 89

Time taken to sort: 0.000000 seconds
```